ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Q1(b) [3 marks]

The smallest resolvable force is 0.122 N.

Q1(c) [3 marks]

$$F = e^(AT) = e^(-0.2) = 0.818731$$

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60^\circ)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$
$$B = [[0], [50]]$$

Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$. With $K_p = 50$, $\theta_d = \pi/2$ and $mg\ell = 2.4525$, solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. This gives θ_{ss} approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19^\circ$, so the gap from 90° is about 2.81°.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

Q3(c) [4 marks]

$$L = [18, 64]^T$$

Q3(d) [2 marks]

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

Q4(c) [4 marks]

With $\dot{\xi}=r-\theta$ and $u=-k_1\theta-k_2\dot{\theta}+K_I\xi$, the homogeneous closed-loop characteristic polynomial is $s^3+(a+bk_2)s^2+(c+$
Desired polynomial: $(s+4)[(s+2)^2+4]=(s+4)(s^2+4s+8)=s^3+8s^2+24s+32$. For a=2,b=1,c=6, matching coefficients gives $k_2=6$, $k_1=18$ and K_I=32.

First multiply $(s+4)(s^2+4s+8) = s^3+8s^2+24s+32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of s.

Q4(d) [2 marks]

Integral action gives zero steady-state error if stable. A practical problem is integrator windup.